

EXERCISE NO.4

1. Write a Java program to create a new array list, add some colors (string) and print out the collection.

Code:

```
import java.util.ArrayList;
import java.util.Iterator;
public class Main {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>(10);

        list.add("Red");
        list.add("Green");
        list.add("Blue");
        list.add("Yellow");
        Iterator<String> iterator = list.iterator();
        while(iterator.hasNext()){
            String data = iterator.next();
            System.out.println(data);
        }
    }
}
```

2. Modify the above Java program to insert an element into the array list at the first position.

Code:

```
import java.util.ArrayList;
public class Main {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>(10);
        list.add("Green");
        list.add("Blue");
        list.add("Yellow");
        System.out.println("List Before Inserting Element at 1st Position in ArrayList:"+list);
        list.add(0,"Red");
        System.out.println("List After Inserting Element at 1st Position in ArrayList:"+list);
    }
}
```

3. Modify the above Java program to retrieve an element (at a specified index) from a given array list.

Code:

```
import java.util.ArrayList;
public class Main {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();
        list.add("Red");
        list.add("Green");
        list.add("Blue");
        list.add("Yellow");
        String element = list.get(2);
        System.out.println("Retrieve Element is :"+element);
    }
}
```

4. Modify the above Java program to update specific array element by given element.

Code:

```
import java.util.ArrayList;
public class Main {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();
        list.add("Red");
        list.add("Green");
        list.add("Blue");
        list.add("Yellow");

        System.out.println("List Before Update:"+list);
        list.set(2, "Black");
        System.out.println("List After Update:"+list);
    }
}
```

5. Modify the above Java program to remove the third element from a array list.

Code:

```
import java.util.ArrayList;
public class Main {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();
        list.add("Red");
        list.add("Green");
        list.add("Blue");
        list.add("Yellow");
        System.out.println("Before Removing third element from Array List:"+list);
        list.remove(2);
        System.out.println("After Removing third element from Array List:"+list);
    }
}
```

6. Modify the above Java program to search an element in a array list.

Code:

```
import java.util.ArrayList;
public class Main {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();
        list.add("Red");
        list.add("Green");
        list.add("Blue");
        list.add("Yellow");
        if (list.contains("Blue")) {
            System.out.println("Element found");
        } else {
            System.out.println("Element not found");
        }
    }
}
```